

Algebra II with Trigonometry

Chapter 5.2

Study guide

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[4401268_alg-2trig-h-chp-5-2-note-docx.pdf](#)

Certainly! Below is a self-contained study guide and summary based on the provided document "Chp 5.2: Trigonometry of Right Triangles."

1. Short Summary

This document introduces the trigonometry of right triangles. It covers the six basic trigonometric ratios, special right triangles (30-60-90 and 45-45-90), and the use of these ratios to solve for unknown sides or angles in right triangles. It further explores special trigonometric values, calculator use, applications (like elevation and depression), and provides step-by-step example problems.

2. Core Definitions, Theorems, and Formulas

Trigonometric Ratios (for angle θ in a right triangle)

- $\sin \theta = \text{opposite} / \text{hypotenuse}$
- $\cos \theta = \text{adjacent} / \text{hypotenuse}$
- $\tan \theta = \text{opposite} / \text{adjacent}$
- $\csc \theta = \text{hypotenuse} / \text{opposite} = 1 / \sin \theta$
- $\sec \theta = \text{hypotenuse} / \text{adjacent} = 1 / \cos \theta$
- $\cot \theta = \text{adjacent} / \text{opposite} = 1 / \tan \theta$

Special Triangles:

45°-45°-90° triangle:

- Legs: 1, 1; Hypotenuse: $\sqrt{2}$

30°-60°-90° triangle:

- Short leg (opposite 30°): 1
- Long leg (opposite 60°): $\sqrt{3}$
- Hypotenuse: 2

Quadrantal Angles:

- 0°, 90°, 180°, 270°, 360° (or 0, $\pi/2$, π , $3\pi/2$, 2π radians)

Special Values (Fill the table):

- For 30° ($\pi/6$), 45° ($\pi/4$), and 60° ($\pi/3$) these values should be memorized.

Angle of Elevation/Depression:

- **Angle of elevation:** Up from the horizontal
- **Angle of depression:** Down from the horizontal

3. Worked Study Guide

Step 1: Understanding the Trigonometric Ratios

Imagine a right triangle with angle θ :  (hypotenuse) p (opposite) o (adjacent side)

- “Opposite” is the side across from θ
- “Adjacent” is next to θ (not the hypotenuse)
- “Hypotenuse” is the longest side (across from the right angle)

Step 2: Using Special Triangles

- **45°-45°-90° triangle:** Both legs are equal, hypotenuse is $\text{leg} \times \sqrt{2}$.
- **30°-60°-90° triangle:** Shortest side (opposite 30°) is half the hypotenuse, side opposite 60° is $\text{hypotenuse} \times \sqrt{3}/2$.

Step 3: Reciprocal Relationships

If you know **sin θ** , you can always find **csc $\theta = 1/(\sin \theta)$** , and the same for other pairs.

Step 4: Using Calculators

- Use mode (degrees/radians) as needed.
- For sec, csc, cot, use reciprocals of base functions.

Step 5: Solving Word Problems

- Draw a diagram.
- Assign sides based on information (heights, distances, angles).
- Use SOH-CAH-TOA or reciprocal.
- For angle of elevation/depression: use $\tan(\theta) = \text{opposite/adjacent}$.

Step 6: Common Exact Trig Values

30°:

- $\sin 30^\circ = 1/2$
- $\cos 30^\circ = \sqrt{3}/2$
- $\tan 30^\circ = 1/\sqrt{3}$

45°:

- $\sin 45^\circ = 1/\sqrt{2} = \sqrt{2}/2$
- $\cos 45^\circ = 1/\sqrt{2} = \sqrt{2}/2$
- $\tan 45^\circ = 1$

60°:

- $\sin 60^\circ = \sqrt{3}/2$
- $\cos 60^\circ = 1/2$
- $\tan 60^\circ = \sqrt{3}$

4. Practice Problems (with Answers)

1. Find the six trigonometric ratios for θ in a right triangle with opposite = 3, adjacent = 4, hypotenuse = 5.

Answer:

- $\sin \theta = 3/5$
 - $\cos \theta = 4/5$
 - $\tan \theta = 3/4$
 - $\csc \theta = 5/3$
 - $\sec \theta = 5/4$
 - $\cot \theta = 4/3$
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2. Given a 30° - 60° - 90° triangle with shortest leg = 5, what is the length of the hypotenuse?

Answer: Hypotenuse = $2 \times 5 = 10$

3. Evaluate $(\cos 30^\circ)^2 - (\sin 30^\circ)^2$ without a calculator.

Answer:

- $\cos 30^\circ = \sqrt{3}/2, \sin 30^\circ = 1/2$
 - $(\sqrt{3}/2)^2 - (1/2)^2 = (3/4) - (1/4) = 1/2$
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4. An observer is 100 ft from a tree. The angle of elevation to the top is 35° . Find the height of the tree.

Answer: Let h = height, $\tan(35^\circ) = h / 100 \Rightarrow h = 100 \times \tan(35^\circ) \approx 70.02$ ft

5. Sketch a right triangle with $\cos \theta = 12/13$. Find $\sin \theta$.

Answer: If $\cos \theta = 12/13$, adjacent = 12, hypotenuse = 13. Therefore, opposite = $\sqrt{(13^2 - 12^2)} = \sqrt{25} = 5$. Thus, $\sin \theta = 5/13$

5. Assumptions & Ambiguities Noted

- The values for the special angles' trigonometric functions were not filled in; I filled those using standard trigonometric values.
 - Some example problems in the document refer to diagrams; I described these as needed.
 - The use of "left-hand trick" is mentioned but not explained in detail. It is likely a mnemonic for the unit circle.
 - For all numeric calculator examples that require rounding, I used standard approximations.
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Let me know if you need more detailed walkthroughs for any specific example or concept!